

LIDGERWOOD, ND, USA

WMO: 727534

Lat: 46.100N

Lon: 97.150W

Elev: 1152

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-12.8	-27.2	1.3	-13.4	-23.3	1.6	-8.7	24.6	30.8	21.8	28.6	3.5	280	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	92.0	71.4	88.5	68.9	85.3	67.1	74.3	88.1	71.9	84.5	69.7	81.9	9.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	113.7	82.9	67.3	104.9	80.3	65.0	96.7	77.7	38.6	88.2	36.4	84.5	34.4	81.8	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.1	16.0	-23.4	98.5	5.1	5.0	-27.1	102.1	-30.1	105.0	-33.0	107.8	-36.7	111.5		
			-20.8	76.8	3.5	3.0	-23.3	79.0	-25.4	80.7	-27.3	82.4	-29.9	84.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	12.2	15.8	28.4	43.8	59.0	67.7	70.1	68.4	58.7	45.1	27.9	16.7	
	DBStd	23.56	13.91	14.58	12.28	10.10	9.71	7.67	6.90	7.07	9.01	9.70	10.26	12.78	
	HDD50	5007	1173	957	669	244	34	0	0	0	23	211	664	1032	
	HDD65	8607	1638	1377	1134	642	231	59	23	45	230	619	1112	1497	
	CDD50	2441	0	0	0	57	315	530	622	571	284	60	2	0	
	CDD65	566	0	0	0	6	47	138	180	151	41	3	0	0	
	CDH74	6413	0	0	0	117	608	1579	2014	1554	469	72	0	0	
	CDH80	2301	0	0	0	34	168	604	807	552	118	18	0	0	
Wind	WSAvg	6.7	7.0	7.1	7.4	7.8	8.1	6.8	5.7	5.3	5.8	6.3	7.0	6.5	
Precipitation	PrecAvg	25.1	0.8	0.6	1.1	2.0	3.1	4.6	3.4	2.4	2.9	2.0	0.9	0.9	
	PrecMax	34.4	2.5	1.2	3.0	4.0	6.9	11.1	7.3	5.6	6.6	6.0	2.8	2.1	
	PrecMin	17.4	0.0	0.1	0.0	0.5	1.0	1.3	1.5	0.7	1.1	0.0	0.0	0.2	
	PrecStd	3.9	0.6	0.3	0.7	1.0	1.4	2.4	1.7	1.2	1.5	1.6	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	51.3	59.0	84.7	89.4	95.0	100.7	95.1	88.4	81.7	59.2	46.7	
		MCWB	37.3	40.9	47.1	60.6	65.1	70.9	74.0	74.6	65.5	60.3	48.9	39.3	
	2%	DB	39.0	45.3	52.0	76.8	84.9	90.5	92.9	89.7	83.2	73.3	51.7	40.7	
		MCWB	33.2	37.8	42.9	56.2	65.1	70.8	72.4	70.1	65.0	55.7	43.7	34.6	
	5%	DB	35.8	39.6	47.7	69.3	80.4	86.9	87.8	85.9	79.4	67.9	48.1	36.9	
		MCWB	31.2	34.4	40.9	53.5	61.6	68.8	69.8	67.4	62.5	53.1	40.8	32.2	
	10%	DB	32.3	35.8	42.6	61.6	77.0	83.0	84.1	82.3	75.4	61.6	43.4	33.6	
		MCWB	28.8	31.5	37.5	48.0	59.3	66.1	67.1	67.0	61.7	50.2	37.4	30.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	42.4	49.0	62.2	69.5	76.7	77.0	76.1	71.2	61.6	51.8	39.6
			MCDB	43.8	50.4	57.5	84.5	85.8	90.2	91.9	88.5	83.2	76.1	57.8	47.4
2%		WB	33.5	38.0	44.3	57.6	66.5	72.8	74.7	73.0	68.2	58.2	45.0	34.9	
		MCDB	38.4	44.9	51.1	74.6	81.2	86.7	89.3	86.4	78.5	71.0	51.4	40.0	
5%		WB	31.5	34.4	41.0	53.4	63.9	70.3	72.4	70.9	65.5	54.7	40.9	32.5	
		MCDB	35.8	39.5	47.2	67.7	77.6	83.7	85.1	82.0	75.9	66.4	47.4	36.6	
10%		WB	29.0	32.0	37.8	49.4	61.3	67.8	69.4	68.6	62.8	50.9	37.5	30.5	
		MCDB	32.3	36.1	42.4	60.7	73.7	80.7	82.4	79.3	73.1	60.9	43.3	33.7	
Mean Daily Temperature Range	5% DB	MDBR	18.1	17.7	16.7	22.2	23.1	22.1	21.7	22.9	24.2	23.5	16.4	16.5	
		MCDBR	20.0	23.1	22.9	33.4	28.2	26.4	25.9	26.4	29.4	33.4	22.9	20.7	
		MCWBR	16.2	18.0	16.5	18.8	14.3	13.0	11.9	13.3	15.6	19.8	16.8	16.4	
	5% WB	MCDBR	20.2	22.7	21.5	31.5	23.8	23.3	22.4	21.9	23.8	29.8	22.0	19.7	
		MCWBR	16.8	17.9	15.9	18.3	13.1	12.4	11.1	12.9	13.6	17.9	16.5	15.8	
Clear-Sky Solar Irradiance	taub	0.273	0.281	0.303	0.355	0.377	0.390	0.385	0.395	0.357	0.320	0.284	0.263		
	taud	2.363	2.380	2.444	2.338	2.351	2.360	2.367	2.331	2.420	2.497	2.501	2.422		
	Ebn at Noon	265	286	295	286	282	278	278	270	272	269	261	258		
	Edh at Noon	23	28	30	37	38	38	37	37	31	25	20	20		
All-Sky Solar Radiation	RadAvg	542	896	1218	1551	1750	1959	2061	1727	1315	826	526	392		
	RadStd	57	96	115	119	120	119	82	152	85	112	58	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page